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Modelling and Analysis of Walk and Intelligent Feeder Service based Last Mile Connectivity to Metro Stations

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Abstract— **Last mile connectivity refers to the critical link that connects commuters from their homes or workplaces to the nearest public transportation hubs, enhancing accessibility and reducing reliance on private vehicles. Existing last mile connectivity solutions to access public transportation often face limitations such as limited coverage areas, inefficient routing and lack of integration with different modes, poor quality of service for commuters, hindering seamless and efficient commuting experience.**

In this paper, an intelligent feeder service is designed to enhance last mile connectivity to metro stations. An online algorithm is developed to recommend suitable boarding junctions for commuters to join shared rides. The algorithm also assists vehicles in navigating the road network by suggesting starting junctions, en route stops, and diversions that maximize the likelihood of picking up commuters, while satisfying the constraints related to the walk distance and additional time required in a shared ride. The performance of proposed system is evaluated for the amount of walking, travel time and waiting time of commuters, and the vehicle occupancy using large scale numeric simulations. The simulation results are encouraging and demonstrate better qualitative performance of the proposed service than other typical feeder or shuttle services.

I. Introduction

Road and rail transportation are the primary modes of transport for moving passengers and goods. In India, the mode share of road and rail transport is 57% and 36%, respectively[1]. The economic growth and urbanization have led to an increased demand for personalized modes of transport in India [2]. The total number of registered motor vehicles has increased from 55 million in 2001 to 292 million in 2021[1] at the growth rate of 10.1% leading to heavy traffic congestion in urban areas. Public transportation plays a vital role in addressing traffic congestion and greenhouse gas emissions. The Indian government has initiated Bus Rapid Transit (BRT) [3] and Metro Rail [4] projects to improve public transportation. However, a major challenge is the lack of first and last mile connectivity. Commuters often face additional travel distances to access public transport stations or return from them. This increases the overall time of their trip. The challenge is exacerbated when the travel to the station is carried out using personal vehicles, as it then requires a parking space near the station.

Efforts are underway in all the countries to enhance the first and last mile connectivity through various means. These include developing feeder bus services, auto-rickshaws, cycle-sharing programs, improving pedestrian infrastructure, and integrating transport modes for seamless transfers. The Netherlands and European countries have many popular bike sharing programs that provides docked/dockless bike access to the residents and visitors in the country [5][6].In India, the initiatives such as park-and-ride facilities, bicycle sharing and provisioning of E-rickshaw at metro stations, and feeder buses have been implemented alongside the Delhi metro rail system [2]. M. Chaturvedi and S. Srivastava [7] propose a generic multi-modal ride sharing framework for last mile connectivity by integrating various modes of transport.

The shuttle rickshaw based paratransit is popular in developing countries like India and Bangladesh. It provides shared rides on small auto-rickshaws for last-mile connectivity. A commuter walks to the nearest junction and waits for an auto-rickshaw to take her to a desired station. An auto-rickshaw starts at a junction, picks up passengers at the junction, and occasionally makes en route stoppages or takes diversions to board more passengers having similar travel requirement. This often results in longer trip times and poor service quality for commuters. In this paper, we propose an intelligent feeder service that maximizes the earning for a driver while ensuring appropriate quality of service for the ride-sharing commuters.

We found many interesting works in the literature on ride sharing. J.F. Cordeau [8] developed a branch-and-cut algorithm to model the Dial-A-Ride Problem (DARP). This algorithm operates offline and assumes prior knowledge of all trip requests. However, finding an exact solution is computationally challenging as it reduces to the NP-hard traveling salesperson problem with time windows. M. Nourinejad et al. [9] designed an agent-based model for dynamic ride sharing, where vehicle owners are willing to share rides with other commuters to reduce travel costs. The model was validated for a one driver - one passenger scenario. Santos and Xavier [10] model the DARP for ride sharing by personal vehicles and taxis with money as incentive. The model of X. Liang et al. [11] considers autonomous taxis and the impact of traffic congestion on ride sharing. All these models serve trips using a single mode and offer door-to-door pickup for each passenger. While this is convenient for passengers, it leads to longer travel times for shared rides and reduces the opportunity for trip sharing. A. Fielbaum [12] designed a ride sharing model where users walk to nearby pick-up and drop-off (PUDO) points. PUDO points are computed

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for each user by solving an NP-hard optimization problem and using a heuristic. The author show that an average walk of one minute reduces the trip rejection rate by over 80% and the Vehicles-Hour-Traveled in more than 10% of cases. W. Zifan [13] observed similar advantage of walking to share a trip. C.F. Daganzo et al. [14] designed a ride-sharing system with premium features such as service time and detour guarantees, and studied its commercial feasibility by analysing the number of rides matched, average passenger trip time, and fleet size requirement for a given demand.

We find that the majority of reported works model the ride sharing problem as an NP-hard problem. They do not leverage the walk flexibility of commuters and do not provide any bounds on additional time spent on a shared trip. Further, none of these papers utilize the historical data of recurrent shared trip requests. In this paper, we address the research gap by proposing an intelligent feeder service. Our novel online algorithm utilizes commuters' walk flexibility and suggests suitable road junctions for shared ride boarding. By analyzing historical trip distribution data, the algorithm determines the attractiveness of junctions for both commuters and vehicles. For a vehicle, the algorithm suggests real-time actions (e.g., wait at the current junction or proceed to a neighboring junction) until it reaches the destination metro station. We evaluate the proposed algorithm using over 9.5 million daily trip requests, considering factors like commuter waiting time, additional time after boarding, and vehicle occupancy during peak and off-peak hours.

The paper is organized as follows: section II presents the proposed system model; section III elaborates on the online algorithm for the intelligent feeder service for ride sharing; the simulation model is detailed in section IV; the simulation results are analyzed in section V; finally, section VI presents the conclusions drawn from the study.

II. System Model

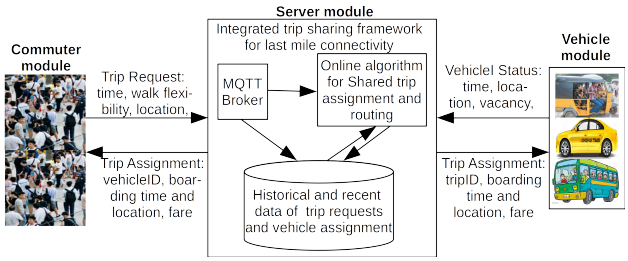


Fig. 1: System model

As shown in Fig. 1, the proposed intelligent feeder service comprises three modules: Commuter module, Vehicle module and Server module. The Commuter module is implemented as a mobile application that allows commuters to send trip requests to server module and receive trip assignments using Message Queuing based Telemetry Transport (MQTT) protocol. The Vehicle module enables vehicles to share their status and receive relevant trip assignments. It can be implemented as an embedded system (consisting of microcontroller, transceivers, and GPS receiver) installed in vehicles or a mobile application installed in the mobile phone of drivers. The Server module, the core component, receives trip requests and vehicle status via an MQTT broker. It includes a database which stores the historical data of trip requests and vehicle assignments. It executes an online algorithm for shared trip assignment and routing of vehicles towards the nearest metro station. The module communicates trip assignments to commuters and vehicles using MQTT protocol. In the following, we describe the functioning of server module.

III. Online Algorithm for Intelligent Feeder Service

This section presents the proposed online algorithm for shared trip assignments and vehicle routing. It is assumed that each commuter aims to reach the nearest metro station, and every trip request includes (time stamp, source location, walk flexibility, additional time flexibility). The walk flexibility determines the maximum distance a commuter is willing to walk for a shared trip. The additional time flexibility specifies the maximum extra time that a commuter is willing to spend after boarding a shared ride compared to a solo ride. This includes on-trip waiting and additional travel time due to detour to board other passengers. It is assumed that every vehicle has the maximum passenger capacity of C. The algorithm is executed in the server module using following inputs: the historical data of spatio-temporal trip generation rate, real-time trip requests from commuters, and the real time vehicle location and occupancy. It suggests to each commuter a junction to walk to for shared ride boarding. The algorithm also suggests action for vehicles at every time step (e.g., wait at current junction, go to a neighbor junction, select the destination metro station when the first commuter boards) until reaching the destination metro station. In the following, we describe how the algorithm computes the suggestions for every commuter module (section III-A) and vehicle module (section III-B) in real time.

A. Computing Response for Commuters

For a new trip request from commuter i, let J^i be the set of reachable junctions based on walk flexibility. For a junction $j \in J^i$, let λ_j^v be the arrival rate of vehicles at j, $d(i,j)$ be the distance between i's location and j, and f_j is the fare to the nearest metro station from j. Accordingly, the utility of commuter i of going to junction j is given by,

$$u_j^i = \lambda_j^v \times \frac{c_1^i}{d(i,j)} \times \frac{c_2^i}{f_j} \tag{1}$$

where c_1^i and c_2^i specify the impact of distance and fare on the commuter, respectively. The probability that the commuter i is assigned to junction j is given by,

$$p_j^i = \frac{u_j^i}{\sum_{j \in J^i} \left(u_j^i\right)} \tag{2}$$

Upon arrival at the suggested junction, the commuter is assigned to the first encountered vehicle. If multiple vehicles are present, the commuter is assigned to the one with more

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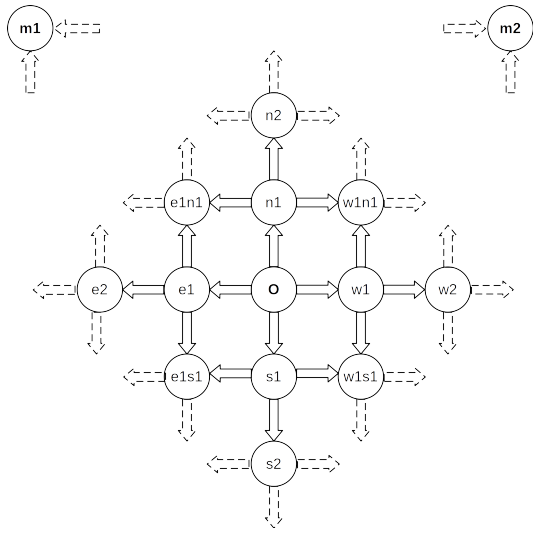


Fig. 2: Example scenario - selection of vehicle route to the nearest metro station

number of passengers with higher probability. This aligns with the common preference of commuters, as they often choose more occupied vehicles for shared trips to minimize en route waiting and detours to board additional passengers.

B. Computing Response for Vehicles

After creation, every vehicle v_k starts maneuver in the road network to find passengers. Let the current location of vehicle be junction o, $NB(o)$ be the one-hop neighbors of o, and $NB^2(o)$ be the two-hop neighbors of o. The probability of suggesting the vehicle to go to junction j is computed based on the relative utility of the options. The utility of junction j is determined by the arrival rate of passengers λ_j^p at j, f_j for going to the nearest metro station from j, and the distance from the vehicle's current location to junction j, $d(o,j)$. It is noted that the exact arrival rate of passengers is known at the current junction, but there is uncertainty for a remote junction due to potential changes in passenger arrival rates. We introduce a damping coefficient $\zeta \leq 1$ and use its appropriate power (ζ^h for h-hop away junction from current location) to account for the uncertainty. Accordingly, the utility of going to an h hop away junction j from o is,

$$u_{o,j}^k = \zeta^h \times \lambda_j^p \times \frac{c_1^k}{d(o,j)} \times c_2^k \times f_j \tag{3}$$

We observe that the utility significantly reduces for the farther away junctions due to distance and damping coefficient. Hence, we consider up to two-hop away junctions in this analysis. The probability that vehicle v_k located at junction o is suggested to go to a junction $j \in \{o\} \cup NB(o) \cup NB^2(o)$ in the next time step is given by,

$$p_{o,j}^k = \frac{u_{o,j}^k}{\sum_{j \in \{o\} \cup NB(o) \cup NB^2(o)} \left(u_{o,j}^k\right)} \tag{4}$$

When the vehicle boards the first passenger at junction o, the algorithm assigns the nearest metro station m_1 as its destination (Fig. 2). Due to the grid road network, multiple routes exist to reach m_1. We use the following probabilistic approach to suggest action to the vehicle at each time step. It is assumed that the vehicle has vacancy to board passenger(s), else it will move on any route (e.g., the shortest route) without stopping at the en route junctions. A vehicle at junction o with available seats may decide to stay at o in the next time step (if additional time flexibility of onboard commuters permits) or move towards one of the junctions: e1, n1, w1, or s1. if the vehicle moves to e_1 or n_1, it progresses towards the destination, while choosing w_1 or s_1 moves it away. To consider this, we evaluate the distance from o to the next junction and the shortest path distance from that junction to the metro station. Let $d(o,j) + d(j,m_1)$ be the minimum distance from o to m_1 through j, where $j \in \{o\} \cup NB(o)$. Further, the vehicle can only divert and move away from the metro station if the onboard passengers additional time flexibility permits. The probability of selecting a junction j is computed using the relative utility of the junction compared to the other options. As earlier, the utility of current junction o and the one-hop away neighbor junction j for going to the metro station m_1 is given by,

$$u_o^{o,m_1} = \frac{\zeta^0 \times \lambda_o^p \times f_o}{d(o,m_1)}; \;\; u_j^{o,m_1} = \frac{\zeta^1 \times \lambda_j^p \times f_j}{d(o,j) + d(j,m_1)} \tag{5}$$

Similarly, the utility of two-hop away junction k (when traveled from o through junction j) for going to m_1 is,

$$u_k^{o,m_1} = \frac{\zeta^2 \times \lambda_k^p \times f_k}{d(o,j) + d(j,k) + d(k,m_1)} \tag{6}$$

As earlier, we use the utility of up to the two-hop neighborhood of o to compute the probability of selecting a junction j for the trip to metro station m. The probability that a vehicle located at junction o goes to junction $i \in \{o\} \cup NB(o)$ during its trip to metro station m is given by,

$$p_{i \in \{o\} \cup NB(o)}^{o,m} = \frac{u_i^{o,m} + \sum_{j \in NB(i)} \left(u_j^{o,m}\right)}{\sum_{i \in \{o\} \cup NB(o)} \left(u_i^{o,m} + \sum_{j \in NB(i)} \left(u_j^{o,m}\right)\right)} \tag{7}$$

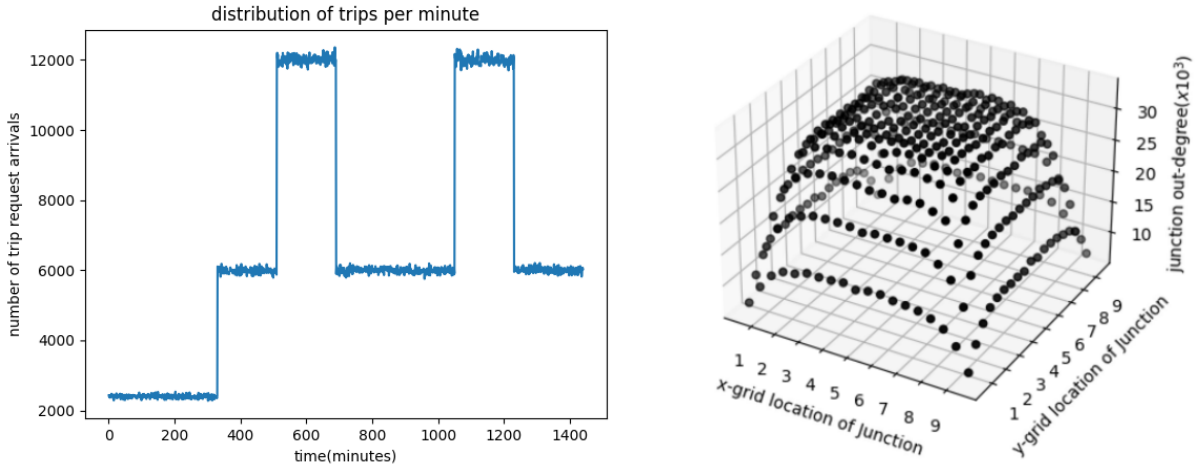
If the vehicle has already exhausted the additional time flexibility of an onboard passenger, it will always move to the next junction towards the destination. Let $NB^m(i)$ be the set of neighbors of junction i such that $d(i,m) > d(j,m), \forall j \in NB^m(i)$. The probability that the vehicle at junction o goes to junction $i \in NB^m(o)$ is,

$$p_{i \in NB^m(o)}^{o,m} = \frac{u_i^{o,m} + \sum_{j \in NB^m(i)} \left(u_j^{o,m}\right)}{\sum_{i \in NB^m(o)} \left(u_i^{o,m} + \sum_{j \in NB^m(i)} \left(u_j^{o,m}\right)\right)} \tag{8}$$

When there are multiple metro stations M^o at the same distance from junction o, the probability of selecting a metro station $m_i \in M^o$ is computed using the utility of up to the two-hop neighborhood of o for a trip to each competing metro station:

$$p_{m_i \in M^o}^o = \frac{\sum_{j \in \{o\} \cup NB(o) \cup NB^2(o)} \left(u_j^{o,m_i}\right)}{\sum_{m_i \in M^o} \sum_{j \in \{o\} \cup NB(o) \cup NB^2(o)} \left(u_j^{o,m_i}\right)} \tag{9}$$

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(a) Distribution of trips in a day (1440 minutes)

(b) Distribution of trips in road network

Fig. 3: Spatio-temporal distribution of trips ($\lambda_{peak} = 30$)

IV. Simulation Setup

In this section, we describe the simulation setup designed for performance analyze of the proposed intelligent feeder service. We use a road network of $10KM \times 10KM$ area wherein a junction is created at every 500 meters, and every grid region is $500m \times 500m$. This leads to a total of 441 (21×21) junctions. It is known that the trip requests are generated at different rates during a day. Given the maximum trip generation rate λ_{peak}, the trips are generated at the rate of $0.2 \times \lambda_{peak}$ for the night duration (midnight to 5:30 AM), $0.5 \times \lambda_{peak}$ for the off-peak duration (5:30 AM to 8:30 AM, 11:30 AM to 5:30 PM, and 8:30 PM to midnight), and at the rate of λ_{peak} for the peak duration (8:30 AM to 11:30 AM and 5:30 PM to 8:30 PM) using Poisson distribution. $\lambda_{peak} = 30\ requests/minute/junction$ is used in the simulations.

Fig. 3a shows the distribution of trips during a day. The x-axis shows the time in minutes, and the y-axis shows the total number of trips generated in the road network during each minute. Cities across the world are classified as the single-centered or multi-centered cities[15]. In a single-center city, there is only one central area that accommodates numerous socio-economic activities and serves as the origin and destination for a large number of trips. In contrast, in a multi-centered city, there are multiple such central areas. In this work, the trips are spatially distributed like a single-centered city area as shown in Fig. 3b. As note earlier, a trip request is specified as a 4-tuple: (time stamp, source location, walk flexibility, additional time flexibility). The walk flexibility distribution is computed using the distance decay function $d = \frac{log_e(1-P)}{\beta}$, where d is distance in miles, $P = e^{-\beta d}$, and $\beta = 1.71$ as suggested by Yang and Diez-Roux [16]. To ensure a reasonable walking distance and accessibility to at least one junction from the source location, we generate P as a random number uniformly distributed in the range of $[0.1, 1.0]$ and add 200 meters to the value of d. The walking speed of each commuter is generated using a Gaussian distribution with mean $\mu = 1.24m/s$ and standard deviation $\sigma = 0.1$[17]. The same available time flexibility of 3 minutes and 5 minutes is used for all the trip requests.

The vehicles with passenger capacity C are generated at a junction using Poisson distribution at the rate of 1/C times the trip generation rate in the region. $C \in \{3, 5\}$ is used in the simulations. The vehicle's maneuver in the road network is determined by the online algorithm described in Section III-B. The vehicle is assumed to travel at a constant speed of $30km/hr$ between selected junctions. The six metro stations are placed at (0,0), (0,2), (0,4), (0,6), (0,8), (0,10) in the $10KM \times 10KM$ grid. The damping factor $\zeta \in \{0.5, 0.6, \ldots, 1.0\}$ is used in the simulations. Table I summarizes the simulation model parameters.

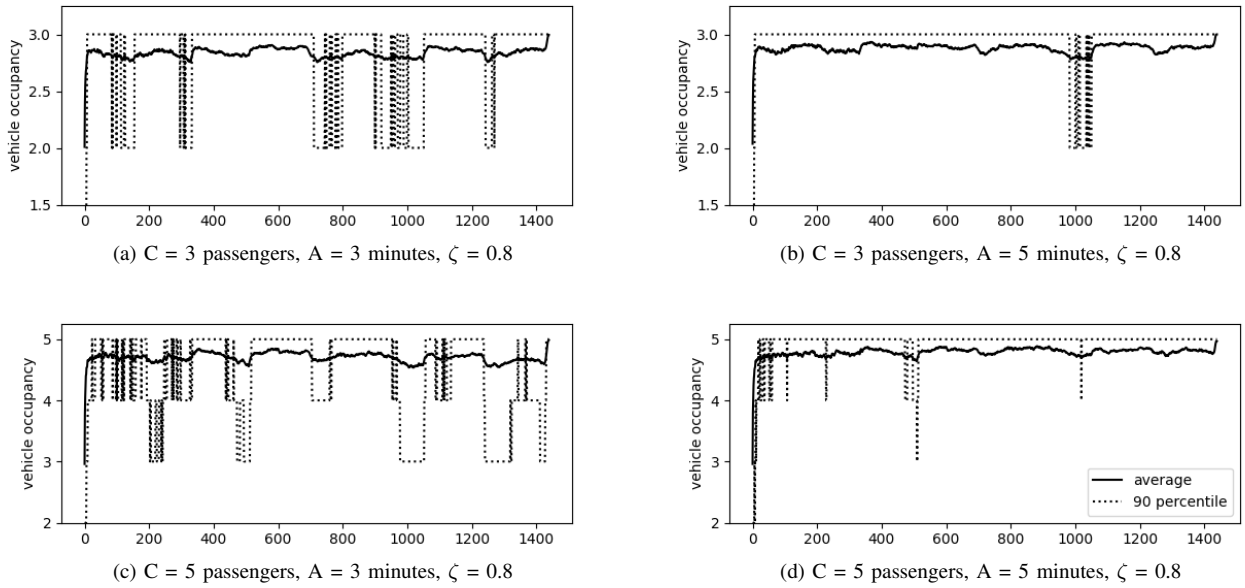
Parameter	Value
Area	$10KM \times 10KM$
Grid size	$500m \times 500m$
Location of metro stations	(0,0), (0,2), (0,4), (0,6), (0,8), (0,10)
λ_{peak}	30 trips/minute/junction
Vehicle capacity C	3, 5
Trip distribution	As mentioned in Fig. 3a and 3b
Walk flexibility	As suggested by Yang et al.[16]
Additional time flexibility A	3, 5 minutes
Walk speed	Gaussian distribution(1.24 m/s, 0.1)[17]
Vehicle distribution	Similar to trip distribution divided by C
Vehicle speed	Constant 30 km/hr when moving
Damping factor ζ	0.5, 0.6, 0.7, 0.8, 0.9, 1.0

TABLE I: Simulation model parameters

V. Simulation Result Analysis

This section presents the performance analysis of proposed intelligent feeder service for last mile connectivity. We assess the performance metrics such as vehicle occupancy, waiting time and additional travel time for commuters in shared rides during both peak hours and off-peak periods.

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(a) C = 3 passengers, A = 3 minutes, ζ = 0.8

(b) C = 3 passengers, A = 5 minutes, ζ = 0.8

(c) C = 5 passengers, A = 3 minutes, ζ = 0.8

(d) C = 5 passengers, A = 5 minutes, ζ = 0.8

Fig. 4: Vehicle occupancy variations during a day (1440 minutes)

a) Vehicle occupancy: Fig. 4 shows the variations in vehicle occupancy during a day (1440 minutes). It is found that the proposed intelligent feeder service achieves the mean vehicle occupancy of more than 2.7 and 4.5 for the vehicle capacity of three passengers and five passengers, respectively. The impact of additional travel time flexibility (A) on vehicle occupancy is studied for $A = \{3, 5\}$ minutes. It is found that the value of $A = 3$ minutes is adequate, but $A = 5$ minutes provides slightly higher average vehicle occupancy and improves the ninety percentile vehicle occupancy during the transition phases of traffic conditions. As can be observed in Fig. 4a and 4c, with $A = 3$ minutes the ninety percentile vehicle occupancy decreases for some time when the transitions in traffic condition occurs from the peak hours to off-peak hours (and vice versa) and the night duration to off-peak hours. Fewer such dips in vehicle occupancy are observed with $A = 5$ minutes. The reason is that with increased additional travel time flexibility, a vehicle can take more stops and diversions to board additional passengers when it is not fully occupied.

b) Additional time for ride sharing compared to solo ride: Fig. 5 shows the percentage of wait time, additional travel time and additional total time required in a shared ride compared to a solo trip. It is observed that more than 75% trip requests (more than 7.2 million out of $\sim$9.5 million trips) have zero waiting time (Fig. 5c). During peak hours, this percentage is more than 80% (more than 3.5 million out of $\sim$4.3 million trips), and for off-peak duration, it is 73% ($\sim$3.8 million out of 5.18 million trips) as shown in Fig. 5a and 5b, respectively. As noted earlier, the additional travel time is upper bounded by the value of $A = \{3, 5\}$ minutes for all the trips.The mean and ninety percentile additional travel time of share rides is 4.45% and 14.29% of the solo trip travel time, respectively. Generally, the last mile trips are short (less than 10 minutes), and the additional travel time turns out to be less than 1.5 minutes. Fig. 5 also shows the percentage of additional total time (sum of wait time and additional travel time) of a shared trip. Similar results are observed for $C = 5$ and $A = \{3, 5\}$ minutes.

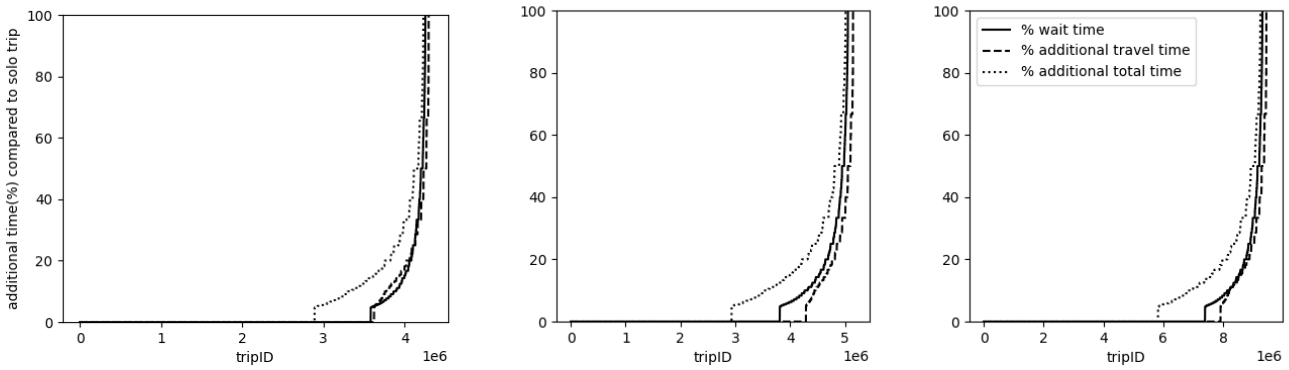
c) Sensitivity analysis of damping coefficient ζ: To study the impact of ζ on the performance metrics, we varied its value from 0.5 (very high damping) to 1.0 (no damping) for C=3 passengers and A=5 minutes. We find that as ζ increases from 0.5 to 1.0, the mean waiting time decreases from 8.97% to 8.13%, from 15.89% to 14.90% and from 12.74% to 11.83%, for peak hours, off-peak duration, and the whole day, respectively. This slight reduction in waiting time with an increase in the value of ζ is attributed to the fact that as ζ approaches 1.0, vehicles move more actively to the next junction to board commuters, rather than waiting at the current junction.

A. Qualitative comparison with other modes:

The proposed intelligent feeder service achieves vehicle occupancy of 2.7 (C=3) and 4.5 (C=5), requiring approximately 35.18 million vehicle trips and 21.11 million vehicle trips, respectively, to serve 95 million commuter requests. If the same number of commuters are served using solo rides with the average vehicle occupancy of 1.1 for two-wheeler vehicles and 1.25 for cars [18], more than 75 million vehicle trips would be required. This could cause significant congestion and greenhouse gas emissions in the region.

The fixed route feeder buses struggle to achieve complete spatio-temporal coverage in a region due to their higher operating costs. The shared shuttle rickshaws are the prevailing mode of transport for last mile connectivity in developing countries. However, they offer poor service quality to commuters due to unpredictable waiting and travel times. The

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(a) Peak duration (b) Off-peak duration (c) Whole day

Fig. 5: Additional time for ride sharing compared to solo ride (C=3 passengers, A=5 minutes, ζ=0.8)

uninformed drivers often take longer stops and diversions to board commuters, resulting in fewer trips per day and reduced earnings. To address these issues, the proposed intelligent feeder service provides bounds on waiting time and additional travel time for commuters, as well as guides vehicles towards more attractive nearby junctions. It also suggests optimal en route stops and detours to facilitate passenger boarding and improve overall service quality.

VI. Conclusion and Future Work

This paper presents an intelligent feeder service aimed at enhancing last mile connectivity to metro stations. An online algorithm is designed that recommends suitable junctions for commuters to board share rides. Additionally, the algorithm assists vehicles in navigating the road network by suggesting starting junctions, en route stops, and diversions to maximize the chances of picking up commuters efficiently. The simulation results demonstrate the effectiveness of the service, with vehicles achieving a mean occupancy of 2.7 (C=3) and 4.5 (C=5). Further, more than 75% commuter requests experience no wait time, and the mean additional travel time compared to a solo trip is less than 15%. This demonstrates effectiveness of the proposed service.

This work can be extended in the following ways. Here we have assumed the steady state of system - there are sufficient vehicles to serve trip demand and there are enough commuters for vehicles. We have done preliminary experiments for the other cases, but a rigorous evaluation is necessary. Further, the system's performance should be assessed for the cases when a fraction of vehicles and commuters do not follow the suggestions provided by the system.

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